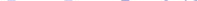


01204211 Discrete Mathematics

Lecture 11b: Context-free languages and grammars (2)¹

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¹Based on lecture notes of *Models of Computation* course by Jeff Erickson. 

Review: Definition

$$\text{DFA } M = (\Sigma, Q, \delta, s, A)$$

$s \in Q, A \subseteq Q$
 $\delta: Q \times \Sigma \rightarrow Q$

A **context-free grammar** consists of the following components:

- ▶ a finite set Σ , a set of symbols (or terminals),
- ▶ a finite set Γ disjoint from Σ , a set of non-terminals (you can think of them as variables),
- ▶ a finite set R of production rules of the form $(A) \rightarrow \underline{w}$ where $A \in \Gamma$ and $\underline{w} \in (\Sigma \cup \Gamma)^*$ is a string of symbols and variable, and
- ▶ a starting non-terminal (usually the non-terminal of the first production rule).

$$(\underline{S}) \rightarrow \underline{1S1} / \underline{0S0} / \underline{0} / \underline{1} / \underline{\varepsilon}$$

Review: Applying the rules

grammar $\rightarrow$ generate string.
 $\searrow$ describe language.
generate. " set of strings.

If you have strings $x, y, z \in (\Sigma \cup \Gamma)^*$ and the production rule

$$(A \rightarrow y,$$

You can apply the rule to the string xAz . This yields the string

$$xyz.$$

We use the notation

$$\underline{xAz \rightsquigarrow xyz}$$

to describe this application.

Review: Derivation

We say that z derives from x if we can obtain z from x by production rule applications, denoted by $x \rightsquigarrow^* z$.

Formally, for any string $x, z \in (\Sigma \cup \Gamma)^*$, we say that $x \rightsquigarrow^* z$ if either

- ▶ $x = z$, or
- ▶ $x \rightsquigarrow y$ and $y \rightsquigarrow^* z$ for some string $y \in (\Sigma \cup \Gamma)^*$.

Review: $L(w)$

The *language* $L(w)$ of string $\underline{w} \in (\underline{\Sigma} \cup \underline{\Gamma})^*$ is the set of all strings in Σ^* that derive from w , i.e.,

$$\underline{L(w)} = \{\underline{x} \in \Sigma^* \mid w \rightsquigarrow^* x\}.$$

The language **generated by** a context-free grammar G , denoted by $L(G)$ is the language of its starting non-terminal.

A language L is **context-free** if there exists some context-free grammar G such that $L(G) = L$.

Review: Parse tree

► 00011

$$S \rightarrow A \mid B$$

$$A \rightarrow 0A \mid 0C$$

$$B \rightarrow B1 \mid C1$$

$$C \rightarrow \varepsilon \mid 0C1$$

Ambiguity

► $1 + 1 + 1 + 1 + 1$

$$S \rightarrow 1 \mid S + S \mid S * S$$

- A string w is **ambiguous** with respect to a grammar G if more than one parse tree for w exists.
- A grammar G is **ambiguous** if some string is ambiguous with respect to G .

More example

$$S \rightarrow 1S1 \mid 0S0 \mid 0 \mid 1 \mid \varepsilon$$

Palindrome in $\{0, 1\}^*$

Consider the following grammar

$$S \longrightarrow 0S1 \mid \varepsilon$$

To show that

$$\underline{L(S) = \{0^n 1^n \mid n \geq 0\}},$$

we have to prove

① $L(S) \subseteq \{0^n 1^n \mid n \geq 0\}$. any string x if $S \xrightarrow{*} x$
 $x = 0^n 1^n$

② $L(S) \supseteq \{0^n 1^n \mid n \geq 0\}$. if $0^n 1^n \rightarrow S \text{ --- } \dots$

Consider the following grammar

$$S \longrightarrow 0S1 \mid \varepsilon$$

To show that

$$L(S) = \{0^n 1^n \mid n \geq 0\},$$

we have to prove

- ✓ ▶ $L(S) \supseteq \{0^n 1^n \mid n \geq 0\}$, and
- ▶ $L(S) \subseteq \{0^n 1^n \mid n \geq 0\}$.

Consider the grammar $S \rightarrow 0S1 \mid \varepsilon$.

$$L(S) \supseteq \{0^n 1^n \mid n \geq 0\}$$

Lemma 1

$S \rightsquigarrow^* 0^n 1^n$ for every non-negative integer n .

Proof.

Consider any non-negative integer n .

Induction Hypothesis: Assume that for every non-negative integer $k < n$, $S \rightsquigarrow^* 0^k 1^k$.

There are two cases to consider.

Case $n=0$: ~~Let us use~~ production rule $S \rightarrow \varepsilon$, $S \rightsquigarrow^* 0^0 1^0 = 0^0 1^0$

Case $n > 0$: ~~on IH.~~ $S \rightsquigarrow^* 0^{n-1} 1^{n-1}$

~~on IH~~ $S \rightsquigarrow 0 \underline{S} 1 \rightsquigarrow^* 0 0^{n-1} 1^{n-1} 1 = 0^n 1^n$

~~also~~ $S \rightsquigarrow^* 0^n 1^n$

Consider the grammar $S \rightarrow 0S1 \mid \varepsilon$.

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Proof.

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Induction Hypothesis: Assume that for every non-negative integer $k < n$, $S \rightsquigarrow^* 0^k 1^k$.

There are two cases to consider.

- ▶ Case 1: $n = 0$.

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Induction Hypothesis: Assume that for every non-negative integer $k < n$, $S \rightsquigarrow^* 0^k 1^k$.

There are two cases to consider.

- ▶ Case 1: $n = 0$.
- ▶ Case 2: $n > 0$.

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Proof.

Consider any non-negative integer n .

Induction Hypothesis: Assume that for every non-negative integer $k < n$, $S \rightsquigarrow^* 0^k 1^k$.

There are two cases to consider.

- ▶ Case 1: $n = 0$.
- ▶ Case 2: $n > 0$. From I.H., we know that

$$S \rightsquigarrow^* 0^{n-1} 1^{n-1}.$$

We can apply rule $S \longrightarrow 0S1$ to obtain $0^n 1^n$, i.e.,

$$S \longrightarrow 0S1 \rightsquigarrow^* 00^{n-1}1^{n-1}1 = 0^n 1^n.$$

In both cases, we conclude that $S \rightsquigarrow^* 0^n 1^n$, as required.



Consider the following grammar

$$S \longrightarrow 0S1 \mid \varepsilon$$

Lemma 2

$$L(S) = \{0^n 1^n \mid n \geq 0\}$$

Proof.

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Consider any string $w \in L(S)$. We show that $w = 0^n 1^n$ for some non-negative integer n .

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Proof.

Consider any string $w \in L(S)$. We show that $w = 0^n 1^n$ for some non-negative integer n .

I.H.: Assume that for any string $x \in L(S)$ such that $|x| < |w|$, $x = 0^k 1^k$ for some non-negative integer k .

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Consider the following grammar

$$S \rightarrow \underbrace{0S1} \mid \underbrace{\varepsilon}$$

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$$L(S) = \{0^n 1^n \mid n \geq 0\}$$

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There are 2 cases:

Case 1: $w = \varepsilon$.

Consider the following grammar

$$S \longrightarrow \underbrace{0S1} \mid \varepsilon$$

Lemma 2

$$L(S) = \{0^n 1^n \mid n \geq 0\}$$

Proof.

Consider any string $w \in L(S)$. We show that $w = 0^n 1^n$ for some non-negative integer n .

I.H.: Assume that for any string $x \in L(S)$ such that $|x| < |w|$, $x = 0^k 1^k$ for some non-negative integer k .

There are 2 cases:

Case 1: $w = \varepsilon$.

Case 2: $w = 0x1$ for some $x \in L(S)$.

Consider the following grammar

$$S \longrightarrow 0S1 \mid \varepsilon$$

w
 $\downarrow$
 x

Lemma 2

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Proof.

Consider any string $w \in L(S)$. We show that $w = 0^n 1^n$ for some non-negative integer n .

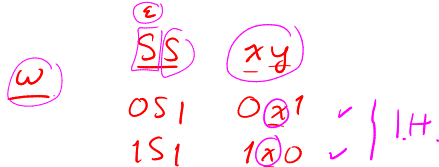
I.H.: Assume that for any string $x \in L(S)$ such that $|x| < |w|$, $x = 0^k 1^k$ for some non-negative integer k .

There are 2 cases:

Case 1: $w = \varepsilon$.

Case 2: $w = 0x1$ for some $x \in L(S)$. Since $|x| = |w| - 2 < |w|$, we can apply I.H., and get that $x = 0^k 1^k$; thus $w = 00^k 1^k 1$, i.e., $w = 0^n 1^n$ where $n = k + 1$, as required. $\square$

Careful



- ▶ When using inductive proof, you have to ensure that each part of the string w is shorter than w .
- ▶ Consider this grammar
$$S \longrightarrow \varepsilon \mid SS \mid \underline{0S1} \mid 1S0.$$
- ▶ When w is created by rule $S \rightarrow SS$, we know that $w = xy$ for $x, y \in L(S)$.
- ▶ Do we know that $|x| < |w|$ and $|y| < |w|$?

Careful

- ▶ When using inductive proof, you have to ensure that each part of the string w is shorter than w .
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$$S \longrightarrow \varepsilon \mid \underline{SS \mid 0S1 \mid 1S0}.$$

- ▶ When w is created by rule $S \rightarrow SS$, we know that $w = xy$ for $x, y \in L(S)$.
- ▶ Do we know that $|x| < |w|$ and $|y| < |w|$?
- ▶ We can consider a minimum-length derivation in the proof to avoid this problem.

$\times$
↪ 7/20/2020.

if $|x| < |w|$ and $|y| < |w|$ then $\#(0, w) = \#(1, w)$
 $\Rightarrow S \xrightarrow{*} w$

Consider grammar $S \rightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0$. For every string $w \in L(S)$, we have $\#(0, w) = \#(1, w)$, where $\#(a, w)$ is the number of occurrences of a in w .

Proof.

Consider $w \in L(S)$. Fix a minimum-length derivation of w .

Induction Hypothesis: Assume that for any string $x \in L(S)$ such that $|x| < |w|$, we have $\#(0, x) = \#(1, x)$.

There are four cases to consider, depending on the first production in this derivation.

- Case 1: The first production is $S \rightarrow \varepsilon$.

Consider grammar $S \rightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0$. For every string $w \in L(S)$, we have $\#(0, w) = \#(1, w)$, where $\#(a, w)$ is the number of occurrences of a in w .

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၁၁၁၁၁ $w = 0x1$ သိသကဲ့သို့ $x \in L(S)$, $|x| < |w|$

၁၁၁၁၁ $I.H.$ $\#(0, x) = \#(1, x)$

၁၁၁၁၁ $\#(0, w) = \#(0, x) + 1 = \#(1, x) + 1 = \#(1, w)$ ✓

Consider grammar $S \rightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0$. For every string $w \in L(S)$, we have $\#(0, w) = \#(1, w)$, where $\#(a, w)$ is the number of occurrences of a in w .

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- Case 2: The first production is $S \rightarrow 0S1$. Case 3: The first production is $S \rightarrow 1S0$.
- Case 4: The first production is $S \rightarrow SS$.

$$w = \underline{xy} \quad \text{if } x \in L(S), y \in L(S)$$

by so on 157 assume
min length derivn,

$$x \neq \varepsilon, y \neq \varepsilon$$

$$\Rightarrow |x| < |w|, |y| < |w|$$

$$\begin{array}{l} \text{X} \quad S \rightarrow SS \rightarrow SSS \rightarrow SSSS \\ \quad \quad \quad \rightarrow \varepsilon SSS \rightarrow \varepsilon \varepsilon SS \rightarrow \varepsilon \varepsilon \varepsilon S \\ \quad \quad \quad \rightarrow \varepsilon \varepsilon \varepsilon \varepsilon = \varepsilon. \end{array}$$

NOT MINIMUM LENGTH

Consider grammar $S \rightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0$. For every string $w \in L(S)$, we have $\#(0, w) = \#(1, w)$, where $\#(a, w)$ is the number of occurrences of a in w .

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Consider grammar $S \rightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0$. For every string $w \in L(S)$, we have $\#(0, w) = \#(1, w)$, where $\#(a, w)$ is the number of occurrences of a in w .

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Consider grammar $S \rightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0$. For every string $w \in L(S)$, we have $\#(0, w) = \#(1, w)$, where $\#(a, w)$ is the number of occurrences of a in w .

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From I.H.,

Consider grammar $S \rightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0$. For every string $w \in L(S)$, we have $\#(0, w) = \#(1, w)$, where $\#(a, w)$ is the number of occurrences of a in w .

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From I.H., we know that $\#(0, x) = \#(1, x)$ and $\#(0, y) = \#(1, y)$; thus,

$$\#(0, w) = \#(0, x) + \#(0, y)$$

Consider grammar $S \rightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0$. For every string $w \in L(S)$, we have $\#(0, w) = \#(1, w)$, where $\#(a, w)$ is the number of occurrences of a in w .

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Consider $w \in L(S)$. Fix a minimum-length derivation of w .

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From I.H., we know that $\#(0, x) = \#(1, x)$ and $\#(0, y) = \#(1, y)$; thus,

$$\begin{aligned}\#(0, w) &= \#(0, x) + \#(0, y) \\ &= \#(1, x) + \#(1, y)\end{aligned}$$

Consider grammar $S \rightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0$. For every string $w \in L(S)$, we have $\#(0, w) = \#(1, w)$, where $\#(a, w)$ is the number of occurrences of a in w .

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From I.H., we know that $\#(0, x) = \#(1, x)$ and $\#(0, y) = \#(1, y)$; thus,

$$\begin{aligned}\#(0, w) &= \#(0, x) + \#(0, y) \\ &= \#(1, x) + \#(1, y) = \#(1, w)\end{aligned}$$

In all cases, we conclude that $\#(0, w) = \#(1, w)$. ✓

Examples: Not palindromes

$$\begin{array}{r} 011001 \\ \hline 0 \end{array} \quad \begin{array}{r} 000110 \\ \hline 1 \end{array}$$

Strings in $(0 + 1)^*$ that are not palindromes.

$$\begin{array}{l} S \rightarrow \underline{0S0} \mid \underline{1S1} \mid \underline{0Z1} \mid \underline{1Z0} \\ Z \rightarrow \underline{\varepsilon} \mid \underline{0Z} \mid \underline{1Z} \end{array}$$

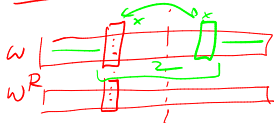
Why does this work?

$$\forall w : S \rightsquigarrow^* w, \\ w \neq w^R$$

$$\bullet L(Z) = \dots$$

$$\bullet L(S) = \dots$$

$$\forall w : w \neq w^R, S \rightsquigarrow^* w$$



Strings with the same number of 0s and 1s

$$\underline{S} \rightarrow \underline{\epsilon} \mid \underline{SS} \mid \underline{0S1} \mid \underline{1S0}.$$

We already show that every string in $L(S)$ contains the same number of 0s and 1s.
Why does it contain all possible required strings?

$$\forall x \in \{0,1\}^*: \quad \underline{\#(0,x) = \#(1,x)} \Rightarrow S \rightsquigarrow^* x.$$

$x = 0100$

• Case 1 x starts with 0 $\rightarrow$ 1

Case 2 x ends with 0 $\leftarrow$ 1

$x = (0)(0)$

$$\begin{aligned} \text{I.H. } \forall y: & \underline{|y| < |x|} \\ & \underline{\#(0,y) = \#(1,y)} \\ & \Rightarrow \underline{S \rightsquigarrow^* y} \end{aligned}$$

Strings in which the number of 0s is greater than or equal to the number of 1s

We can start with the previous grammar

$$\underline{S \longrightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0.}$$

And try to add more rules.

Strings in which the number of 0s is greater than or equal to the number of 1s

We can start with the previous grammar

$$S \longrightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0.$$

And try to add more rules.

$$S \longrightarrow \varepsilon \mid SS \mid 0S1 \mid 1S0 \mid \underline{0S} \mid \underline{S0}.$$

Strings with different numbers of 0s and 1s

We can start with the previous grammar E of strings with equal number of 0 and 1.

$$E \longrightarrow \varepsilon \mid EE \mid 0E1 \mid 1E0.$$

There are two cases.

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Strings with different numbers of 0s and 1s

We can start with the previous grammar E of strings with equal number of 0 and 1.

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There are two cases.

$$S \longrightarrow O \mid I$$

$$O \longrightarrow E0O \mid E0E$$

How about I ?

Strings with different numbers of 0s and 1s

We can start with the previous grammar E of strings with equal number of 0 and 1.

$$E \longrightarrow \varepsilon \mid EE \mid 0E1 \mid 1E0.$$

There are two cases.

$$S \longrightarrow O \mid I$$

$$O \longrightarrow E0O \mid E0E$$

How about I ?

$$I \longrightarrow E1I \mid E1E$$

Balanced parentheses

$$S \longrightarrow (S) \mid SS \mid \varepsilon$$

Balanced parentheses

$$S \longrightarrow (S) \mid SS \mid \varepsilon$$

$$S \longrightarrow (S)S \mid \varepsilon$$

Mutual induction

Consider grammar

$$S \longrightarrow 0A1 \mid \varepsilon$$

$$A \longrightarrow 1S0 \mid \varepsilon$$

What is $L(S)$?

Mutual induction

Consider grammar

$$S \longrightarrow 0A1 \mid \varepsilon \qquad A \longrightarrow 1S0 \mid \varepsilon$$

What is $L(S)$?

From inspection, we may guess that $L(S) = (01)^*$. But how can we prove that?

Mutual induction

Consider grammar

$$S \longrightarrow 0A1 \mid \varepsilon \qquad A \longrightarrow 1S0 \mid \varepsilon$$

What is $L(S)$?

From inspection, we may guess that $L(S) = (01)^*$. But how can we prove that?

To prove $L(S) = (01)^*$, we must also prove $L(A) = (10)^*$ *at the same time*.